

MTH-202 Discrete Mathematics Update MCQS For Mid Term Solve By Vu Topper RM

85% To 100% Marks



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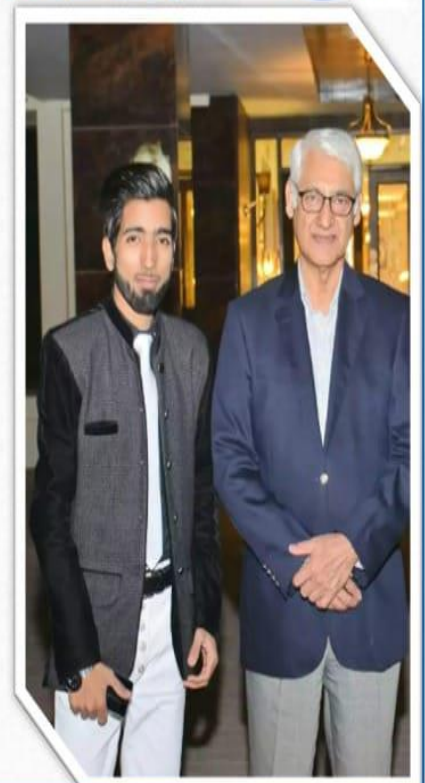
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Range of function $f(x) = \{e^x\}$ is.....

- A. Set of integers
- B. Set of negative Real number.
- C. Set of positive Real numbers**
- D. Set of Real number

The logic gate NOT is a uniary operation on $\{0,1\}$.

- A. false
- B. true**

If $f^{-1}(x)=6-x^2$ then $f^{-1}(2)$ is

- A. 2**
- B. 1
- C. 0
- D. 3

Given a Set $A=\{a,b\}$ and $B=\{1,2,3\}$, then number of functions from A to B is

- A. 6
- B. 8
- C. 5
- D. 9**

How many one-to-one functions are there from a set with five elements to a set with six elements.

- A. 3 factorial (3!)
- B. 4 factorial(4!)
- C. 6 factorial(6!)**
- D. 5 factorial(5!)

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- A. true**

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B. false

Which is not a binary operation on the set of natural numbers N ?

A. Multiplication

B. Subtraction

C. None of these.

D. Addition

The relation $R = \{(12, 14), (13, 5), (-2, 7), (x, 13)\}$ is a function if x is not equal to

A. 14, 7 or 13.

B. 12, 13 or 2.

C. 12, 13 or -2.

D. 14, 5 or 7.

Let $f(x) = x$ and $g(x) = -x$ for all $x \in \mathbb{R}$, then $(f+g)(x)$ is _____.

A. $-x$

B. 0

C. $2x$

D. $-2x$

The Range of $f: X \rightarrow Y$ is also called the image of X under

A. true

B. false

Which of the following relations are functions? $R_1 = \{(3, 4), (4, 5), (6, 7), (8, 9)\}$, $R_2 = \{(3, 4), (4, 5), (6, 7), (3, 9)\}$, $R_3 = \{(-3, 4), (4, -5), (0, 0), (8, 9)\}$, $R_4 = \{(8, 11), (34, 5), (6, 17), (8, 19)\}$

A. R_3 and R_2 are functions

B. R_1 and R_3 are functions

C. R_1 and R_2 are functions

D. R_2 and R_4 are functions

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Cardinality of positive prime numbers less than 20 is

- A. 5
- B. 10
- C. 8**
- D. 20

Which of the following is not a binary operation on the set of Integers?

- A. Multiplication
- B. Subtraction
- C. Addition
- D. Division**

A set is countably infinite if, and only if, it has the same cardinality as the set of

- A. whole numbers
- B. real numbers
- C. positive integers**
- D. integers

Let f be a function from $X = \{2, 4, 5\}$ to $Y = \{1, 2, 4, 6\}$ defined as $f = \{(2, 6), (4, 2), (5, 1)\}$. Then the range of f is _____ .

- A. $\{2, 4, 5\}$
- B. $\{1, 2, 6\}$**
- C. $\{1, 4, 6\}$
- D. $\{1, 2, 4, 6\}$

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x - \sqrt{x}$ is a real valued function.

- A. true
- B. false**

A graph of a function f is onto iff every horizontal line intersects the graph in at least one point.

- A. true**

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B. false

Given a set X , define a function i from X to X by $i(x) = x$ from all x belonging to X . Then _____.

A. i is both injective and surjective

B. i is injective but not surjective

C. i is surjective but not injective

D. i is neither injective nor surjective

A constant function is one to one iff its is a singleton.

A. Range

B. Domain

If $f^{-1}(x) = 6 - x^2$ then $f^{-1}(2)$ is

A. 2

B. 3

C. 1

D. 0

The logic gates OR and AND are unary operation on $\{0,1\}$.

A. true

B. false

Let $f(x) = x + 3$ then $f^{-1}(x)$ is

A. $x + 3$

B. $x - 3$

C. $3 - x$

D. $2x - 3$

The range of a function is always the co-domain of f .

A. a superset of

B. equal to

C. a subset of

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D. not equal to

The number of elements in $A \times B$ are if A is a set with '5' elements and B is a set with '4' elements.

A. 0

B. 20

C. 1

D. 9

A function whose range consists of only one element is called.....

A. One to one function

B. Onto function

C. Constant function

D. Identity function

A set is countable if, and only if, it is _____.

A. infinite

B. finite or countably infinite

C. finite

D. countably infinite

If $f(x)=2x$ and $g(x)=x$ then $g(f(x))$ is.....

A. $2x$

B. $2x^2$

C. x

D. x^2

Let $X = \{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1) = 1$, $f(2) = 1$, $f(3) = 1$, $f(4) = 1$ then which of the following is true?

A. f is an identity function

B. f is injective

C. f is a constant function

D. f is bijective

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Domain of the relation $\{(0,1),(3,22),(90,34)\}$ is _____ .

- A. $\{1,22,34\}$
- B. $\{0,1,3\}$
- C. $\{0,1,3,22,90,34\}$
- D. $\{0,3,90\}$**

Let $P(A)$ be the power set of a non-empty set A and by using the definition of “subset” relation, $\subseteq$, for all $X, Y \in P(A)$, $X \subseteq Y \Leftrightarrow \forall x$, iff $x \in X$ then $x \in Y$. Then _____.

- A. $\subseteq$ is a partial order relation.**
- B. $\subseteq$ is not a partial order relation.

Let R be the universal relation on a set A then which one of the following statement about R is true?

- A. R is not reflexive
- B. R is not transitive
- C. R is reflexive, symmetric and transitive.**
- D. R is not symmetric

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1)(2, 2)(3, 3)(4, 4)\}$ then R is

- A. Symmetric
- B. Options a,b,and c all true
- C. Reflexive**
- D. Transitive

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is.....

- A. $\{(a,1)(2,b)(3,c)(4,d)\}$
- B. $\{(1,a)(2,b)(3,c)(4,d)\}$**
- C. $\{(a,1)(b,2)(3,c)(4,d)\}$
- D. $\{(1,a)(b,2)(3,c)(4,d)\}$

Range of the relation $\{(0,1),(3,22),(90,34)\}$ is _____ .

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- A. {0,3,90}
- B. {0,1,3,22,90,34}
- C. {1,22,34}**
- D. {0,1,3}

Let $A = \{1,2,3\}$ and R be the relation defined on A such that $R = \{(1,1), (1,2), (2,3), (3,1)\}$, then R is _____ .

- A. Symmetric
- B. Antisymmetric**

Let R and S be transitive relations on a set A then _____

- A. Both R union S and R intersection S are transitive
- B. R intersection S is transitive**
- C. Neither R union S is transitive nor R intersection S is transitive
- D. R union S is transitive

There is atleast one loop in the graph of an irreflexive relation.

- A. true
- B. false**

Let R be a binary relation on a set A . R is anti-symmetric iff _____.

- A. $a, b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a = b$
- B. $a, b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a \neq b$**
- C. $a, b \notin A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a \neq b$
- D. None of the above

Find x and y given $(2x, x + y) = (6, 2)$.

- A. $x=2, y=1$
- B. $x=3, y=-1$**
- C. $x=2, y=-1$
- D. $x=3, y=1$ $x=2, y=1$

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Domain of a relation symbolically written as _____.

A. None of these

B. $\text{Dom}(R) = \{a \in A \mid (a, b) \in R\}$

C. $\text{Dom}(R) = \{b \in B \mid (a, b) \in R\}$

D. $\text{Dom}(R) = \{a \notin A \mid (a, b) \in R\}$

Let $A = \{0,1\}$ and $B = \{1\}$. Let R and S be two binary relations on Cartesian product of A and B such that $R = \{(0,1)\}$ and $S = \{(1,1)\}$. Then $R \cap S =$ _____.

A. $\{1,1\}$

B. $\{(0,1)\}$

C. $\{0,1\}$

D. empty

If $x \equiv -10 \pmod{15}$ Which of the following integers are valid solution for x ?

A. 7

B. 5

C. 3

D. 6

Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Range of the inverse of the relation is _____.

A. $\{1, 3, 5, 7\}$

B. $\{5, 6, 7, 8\}$

C. $\{1, 2, 3, 4\}$

D. $\{2, 4, 6, 8\}$

Let R be a relation on a set A . If R is symmetric then its complement is _____.

A. Irreflexive

B. Symmetric

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C. Reflexive

D. Antisymmetric

If $A = \{1, 2, 3\}$ is a set and $R = \{(1, 2), (2, 2), (2, 1)\}$ is a relation on A , R is

A. Reflexive

B. Symmetric

C. None of these

D. Transitive

If a relation $R = \{(1,1)(2,1)(1,2)(2,2)\}$ is given then which of the following is not true about this relation.

A. R is Symmetric.

B. R is Transitive

C. R is Reflexive.

D. R is Irreflexive.

If $x \equiv 17 \pmod{5}$ Which of the following integers are valid solution for x ?

A. 4

B. -44

C. 8

D. 12

Let R be a relation on a set A . If R is reflexive then its compliment is

A. Reflexive

B. Antisymmetric

C. Symmetric

D. Irreflexive

Let the set $A = \{1, 2, 3, 4\}$ Then the relation $\{(2, 4), (4, 2)\}$ is _____.

A. Transitive

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- B. AntiSymmetric
- C. Reflexive
- D. Symmetric**

The union of the transitive binary relations $R = \{(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)\}$ and $S = \{(2,1), (2,4), (3,3)\}$ on the set $A = \{1,2,3,4\}$ is given by $R \cup S = \{(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)\}$ then $R \cup S$ is_____?

- A. not transitive**
- B. transitive

Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Domain of the inverse of the relation is _____.

- A. {2, 4, 6, 8}**
- B. {1, 2, 3, 4}
- C. {1, 3, 5, 7}
- D. {5, 6, 7, 8}

Let $A = \{1,2,3,4\}$ and define the relation R on A by $R = \{(1,2), (2,3), (3,3), (3,4)\}$. Then _____

- A. R is neither reflexive nor irreflexive**
- B. R is both reflexive and irreflexive
- C. R is reflexive
- D. R is irreflexive

Range of a relation symbolically written as _____.

- A. None of the above
- B. $\text{Ran}(R) = \{a \in A \mid (a, b) \in R\}$
- C. $\text{Ran}(R) = \{b \in B \mid (a, b) \in R\}$**
- D. $\text{Ran}(R) = \{b \notin B \mid (a, b) \in R\}$

The matrix representation of an inverse relation can be obtained by taking theof the relational matrix.

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A. transpose

B. Determinant

C. inverse

D. adjoint

Let R be a relation on a set A . R is transitive if and only if for all $a, b, c \in A$ then:

A. $(a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$

B. $(a, b) \in R$ then $(b, a) \notin R$

C. $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$

D. $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ and $(a, b) \in R$ then $(b, a) \in R$

The properties of being symmetric and being anti-symmetric are _____.

A. Negative of each other

B. Not negative of each other

Let $A = \{1, 2, 3, 4\}$ and define a relation R on A by $R = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$. Then which one of the following is a correct statement about R :

A. R is both transitive and reflexive

B. R is neither reflexive nor transitive

C. R is transitive

D. R is reflexive

Let $A = \{0, 1, 2\}$ and $R = \{(0, 2), (1, 1), (2, 0)\}$ be a relation on A . Then which of the following statement about R is true?

A. R is reflexive

B. R is symmetric

C. None of the above

D. R is transitive

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- A. Matrix has always 0 in its diagonal entries
- B. Matrix has always 1 in its diagonal entries
- C. Matrix is singular
- D. Matrix is equal to its transpose.**

- A. $A \times B$
- B. $A \times B$ (intersection) R
- C. $A \times B$ (difference or -) R**
- D. $A \times B$ (Union) R

A. (3,3)

- Set of positive real numbers

- $\text{SoR} = \{(a, c) \in A, c \in C, 3e \in B, (a, b) \in R \text{ and } (b, c) \in S\}$

- 12

- $\{1, 22, 34\}$

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Let $A = \{0,1,2\}$ and $R = \{(0,2),(1,1),(2,0)\}$ be a relation on A . The which of the following ordered pairs are needed to make it transitive?

- $(0,0)$ and $(2,2)$

Operation of subtraction is a binary operation on the set of _____

- Integers

Let $S=R$ and define the 'square' relation $R = \{(x,y) | x^2=y^2\}$. The square relation is an _____ relation

- Equivalence relation

The logic gate NOT is a unary operation on $\{0,1\}$.

- True

Let $A = \{1,2\}$, then $P(A) =$ _____

- $\{\{\},\{1\},\{2\},\{1,2\}\}$

If a relation R is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

- Inverse relation will be irreflexive.

Let R be a binary relation on a set A , R is anti-symmetric iff _____

- $a,b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a=b$

"-" is a binary operation on the set of integers Z .

- True

The inverse relation R^{-1} from B to A is defined as _____

- $R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$

Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix is equal to its transpose

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Let $A = \{1, 2, 3, 4\}$ and let R and S be transitive binary relations on A defined as; $R = \{(1, 2), (1, 3), (2, 2), (3, 3), (4, 2), (4, 3)\}$ and $S = \{(2, 1), (2, 4), (3, 3)\}$ then $R \cup S = \{(1, 2), (1, 3), (2, 1), (2, 2), (2, 4), (3, 3), (4, 2), (4, 3)\}$

- R union S is transitive

Let $S = R$ and define the 'square' relation $R = \{(x, y) | x^2 = y^2\}$. The square is an _____ relation.

- Equivalence relation

If $x \equiv -10 \pmod{15}$. Which of the following integers are valid solution for x ?

- 5

Let R be a binary relation on a set A . If R is anti symmetric then _____

- Inverse of R is anti symmetric

If $A = \{1, 2, 3\}$ is a set and $R = \{(1, 2), (2, 2), (2, 1)\}$ is a relation on A , R is

- Symmetric

Let $A = \{0, 1\}$ and $B = \{1\}$. Let R and S be two binary relations on Cartesian product of A and B such that $R = \{(0, 1)\}$ and $S = \{(1, 1)\}$. Then $R \cap S =$ _____

- Empty

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- Partial order Relation

Let $X = \{1, 2, 3\}$ and $Y = \{7, 8, 9\}$ and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- Co-domain of f must contain 1 element

The function fog and gof are always equal

- False

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If a relation $R = \{(1,2), (2,3), (3,4), (4,1), (2,2)\}$ is given then which of the following is true about this relation.

- R is reflexive

A set is called countable if , and only if, it is _____

- finite

Let $f(x) = x^2 - 1$ define function f from R to R and $c=2$ be any scalar, then $c, f(x)$ is _____

- $2x^2 + 2$

The set Z of all integers is _____

- Countable

Let R be a binary relation on a set A . If R is anti symmetric then _____

- Inverse of R is symmetric

For $(2x-3, 4y+2) = (1, 10)$. What will be the value of x and y ?

- $(2, 2)$

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root of } x^2$ for all $x \in R$. Then _____

- $F(x)$ is not equal to $g(x)$

If a set A has 15 elements then $P(A)$ (power set of A) has _____ elements.

- 2^{15}

For the relation below to be a function, x cannot be what values $\{(12, 14), (13, 5), (-2, 7), (x, 13)\}$?

- x cannot be 12, 13, or -2

Let the set $A = \{1, 2, 3, 4\}$. Then the relation $\{(2, 4), (4, 2)\}$ is _____

- Symmetric

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For the following relation to be a function, x can not be what values? $R = \{(2,4), (x,1), (4,2), (5,6)\}$.

- **x cannot be 2,4 and 5**

Vertical line test is used to determine that whether the graph of a relation is a function or not.

- **True**

The properties of being symmetric and being anti symmetric are

- Not negative of each other

The number of elements in $A \times B$ are _____ if A is a set with '5' elements and B is a set with '4' elements.

- 20

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is _____

- $\{(1,a)(2,b)(3,c)(4,d)\}$

Logic gate NOT does not define a binary operation on $(0,1)$ because

- It takes a single input and gives a single output

How many real numbers exist between 1 and 5

- 3

The number pi is

- Irrational

The number square root 2 is

- Irrational

Range the relation $\{(0,1)(3,22),(90,34)\}$

- $\{0, 3, 90\}$

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Spherical coordinate 0 is related to the cylindrical coordinate as _____

- $@=@$

Operation of subtraction is a binary operation on the set _____

- Integers

Let $A = \{1,2,3,4\}$ and $R = \{(1,2), (2,3), (3,3), (3,4)\}$ be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

- (3,3)

Input values of the function are called the _____

- Domain

Range of function $f(x) = |x|$ will be

- Set of positive real numbers

Which of the following is not a binary operation on the set of integers?

- Division

In the matrix representation of an irreflexive relation all the entries in the main diagonal are _____

- 0

If the partition set of A is $\{A_1, A_2\}$ then

- $A_1 \cap A_2 = \text{not empty set}$

Let $A = \{a, b\}$ then $P(A) =$

- $\{\text{Non empty set}, \{a\}, \{b\}, \{a, b\}\}$

Which relations below are not functions?

- $\{(13,14), (13,5), (16,7), (18,13)\}$

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In the directed graph of an antisymmetric relation there is _____ pair of arrows between two distinct elements of the set.

- **No**

If a relation $R = \{(1,1), (2,1), (2,2)\}$ is given then which of the following is not true about this relation

- **R is irreflexive**

Let R and S be transitive relations on a set A then _____

- Neither $R \cup S$ is transitive nor $R \cap S$ is transitive

Let $R = \{(1,2)(3,4)(5,6)(7,8)\}$. Domain of the inverse of the relation is

- $\{2, 4, 6, 8\}$

Let $A=\{1,2,3,4,5\}$ and $B=\{4,9,16,17,25\}$. Then the relation $R=\{(2,4),(3,9),(4,16),(9,17)\}$ The inverse of R is'

- $\{(4,2), (9,3), (16,4), (17,3)\}$

Let R be a relation on a set A . If R is symmetric then its complement is

- Irreflexive

Which is not a binary operation on the set of natural numbers $\mathbb{N}$?

- **Subtraction**

If a relation $R=\{(1,1)(2,1)(1,2)(2,2)\}$ is given then which of the following is not true about this relation.

- **R is irreflexive**

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is

- $\{(1,a)(2,b)(3,c)(4,d)\}$

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For any set A, the Cartesian product of A and A is known as _____

- Universal relation

Let $A=\{p,q,r,s\}$ and define a relation R on A by $R=\{(p,p),(p,r),(q,r),(q,s),(r,s)\}$ Then which one of the following is the correct statement about R:

- R is not reflexive

$A=\{1,2\}$ $B=\{3,4\}$, $R=\{(1,3)(2,4)\}$. Then the complement of R is _____.

- $\{(1,4)(2,3)\}$

Domain of a relation symbolically written as _____.

- $\text{Dom}(R)=\{a \in A \mid (a,b) \in R\}$

Let $X=\{2,4,5\}$ and $Y=\{1,2,4\}$ and R be a relation from X to Y defined by $R=\{(2,4)(4,1)(a,2)\}$. For what value of 'a' the relation R is a function?

- 5

Let $A=\{1,2,3,4,5,6,7,8,9\}$, then which of the following sets represent the partition of the set A?

- $A=\{1,3,5,7,9\}$, $B=\{2,4,6\}$, $C=\{8\}$

Let $A=\{1,2,3\}$ and $B=\{2,4\}$ then number of binary relations from A to B are _____.

- 64

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- Partial order Relation

Let f be a function from $X=\{2,4,5\}$ to $Y=\{1,2,4,6\}$ defined as: $f=\{(2,6),(4,2),(5,1)\}$. The range of f is _____

- $\{1,2,6\}$

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Let $A=\{0,1,2\}$ and $R=\{(0,2),(1,1),(2,0)\}$ be a relation on A. Then which of the following statement about R is true?

- R is symmetric

Let $A=\{2,3,4\}$ and $B=\{2,6,8\}$ and let R be the "divides" relation from A to B i.e for all (a,b) belong to (Cartesian product of A and B), $a,R b$ iff $a \mid b$ (a divides b). Then

- $R=\{(2,2),(2,6),(2,8),(3,6),(4,8)\}$

Let $A=\{1,2,3,\dots,50\}$ and $B=\{2,4,6,8,10\}$. Then the Cartesian product of A and B has _____ elements.

- 250

In the matrix representation of an reflexive relation all the entries in the main diagonal are _____

- 1

Which of the following is not a type of a relation?

- Permutation

Let $X=\{2,4,5\}$ and $Y=\{1,2,4\}$ and R be a relation from X to Y defined by $R=\{(2,4),(4,1),(a,2)\}$. For what value of 'a' the relation R is a function ?

- 5

Which of the following is not a representation of a relation?

- Venn diagram

Let $A=\{1,2,3,4\}$ and define the following relations on A. Then $R=\{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$ is _____

- R is irreflexive

The range of $f:X \rightarrow Y$ is also called the image of

- True

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Complementary Relation symbolically written as _____

- $R = A * B - R = \{(a,b) \in A * B \mid (a,b) \text{ not belong to } R\}$

Let $A = \{1,2,3,4\}$ and $R = (1,1)(2,2),(3,3),(4,4)$ then R is

- All options

If a relation $R = \{(1,1),(2,1),(1,2),(2,2)\}$ is given then which of the following is not true about this relation.

- R is irreflexive

Which of the following logical connective is not a binary operation ?

- Implication

A set may be dividend up into its disjoint subsets, such division is called _____

- Partition

If $A = (1,2,3) \& B = (4,5,6)$ and $R = \{(1,4)(2,5)(3,6)(3,4)\}$ The complementary relation is _____

- $A * B$ (difference or -) R

In matrix representation of a _____ relation, the diagonal entries are always 1.

- Reflexive

R is not symmetric iff there are elements a and b in A such that _____

- (a,b) belongs to R but (b,a) does not belong to R

Which relations below are functions?

$R_1 = \{(3,4),(4,5),(6,7),(8,9)\}$

$R_2 = \{(3,4),(4,5),(6,7),(3,9)\}$

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$R_3 = \{(-3,4), (4,-5), (0,0), (8,9)\}$

$R_4 = \{(8,11), (34,5), (6,17), (8,19)\}$

- R_1 and R_3 are functions

The logic gate OR and AND are unary operation on $\{0,1\}$

- False

There is atleast one loop in the graph of an irreflexive relation

- False

There is atleast one loop in the graph of an reflexive relation

- True

A contains 3 elements and B contains 2 elements, then number of subsets of $A \times B$ are _____

- 64

Let $A = \{1,2,3\}$ and $B = \{0,1,2\}$ and $C = \{a,b\}$ $R = \{(1,0), (1,2), (3,1), (3,2)\}$
 $S = \{(0,b), (1,a), (2,b)\}$ composite of R and S = _____

- $\{(1,b), (1,a), (3,a), (3,b)\}$

If R is transitive then the inverse relation will be transitive

- True

The number of elements in the power set of P(not empty set) denoted by $P(P(\text{not empty set}))$ is

- 2

The function defined from Z to Z as $f(x) = 1/(x+2)(x-2)$ is not well defined because _____

- Function is not defined at $x=2$ and $x=-2$

Rang of relation $\{(0,1), (3,22), (90,34)\}$ is _____

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- $\{1, 22, 34\}$

The number of elements in the power set of P(not empty set) denoted by P(not empty set) is

- 1

Let $R = \{(1,2), (3,4), (5,6), (7,8)\}$. Domain of inverse of the relation is _____

- $\{2,4,6,8\}$

The relation 'divides' on the set of integers is _____

- A symmetric relation

Operation of subtraction is a binary operation on the set of _____

- Integers

If R is transitive then the inverse relation will be transitive.

- True

Let $A = \{1,2,3,4\}$ and define the relation R on A by $R = \{(1,2), (2,3), (3,3), (3,4)\}$. Then _____

- R is both reflexive and irreflexive

A set may be divided up into its disjoint subsets, such division is called

- Partition

If a set A contains n elements then the number of elements in its power set P(A) is _____

- 2^n

Range of a relation symbolically written as _____.

- $\text{Ran } R = \{b \in B \mid (a, b) \in R\}$

Let R be a binary relation on a set A. If R is anti symmetric then _____.

- Inverse of R is anti symmetric

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Let A be a set with m elements and B be a set with n elements then the number of elements in $A \times B$ are _____

- m.n

Let $A = \{1, 2, 3, 4\}$ and define the following relation on A. Then $R = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 2)\}$ is _____

- R is symmetric

If $A = \{1, 2, 3\}$ & $B = \{4, 5, 6\}$ and $R = \{1, 4\}, \{2, 5\}, \{3, 6\}, \{3, 4\}$. The complementary relation is _____

- $A \setminus B$ (difference or -) R

Let R be a relation on a set A. If R is reflexive then its complement is _____

- Irreflexive

$25 \equiv 1 \pmod{3}$ means that 3 divides _____

- 25-1

Let R be the universal relation on a set A then which one of the following statement about R is true ?

- R is reflexive, symmetric and transitive

Domain of a relation symbolically written as _____

- $\text{Dom}(R) = \{a \in R \mid (a, b) \in R\}$

Let R be a relation on a set A. R is transitive if and only if for all $a, b, c \in A$ then

- $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$

Let A be a non-empty set and $P(A)$ the power set of A Define the 'subset' relation, $\subseteq$ as follows for all $X, Y \in P(A)$, $X \subseteq Y \iff$ for all x , iff $x \in X$ then $x \in Y$. Then $\subseteq$ is _____

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- c is partial order relation

Define a relation $R = \{(1,1), (2,2), (3,3), (1,3)\}$ the relation is

- R is reflexive and transitive

Let R be a relation on a set A . R is transitive if and only if for all $a, b, c \in A$ then.

- $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

Let R and S be reflexive relations on a set A then $R \cap S$ is reflexive

- True

A function whose range consists of only one element is called _____

- One to one function

The set Z of all integers is _____

- Countable

One-to-one correspondence means the condition of _____

- Both (a) and (c)

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$, Define a function f from X to Y such that $f(1) = 7, f(5) = 3, f(9) = \underline{\hspace{2cm}}$. Which is true $f(9)$ to make it a one-to-one (injective) function?

- 4

What will be the fourth term of the following sequence $1/2, 2/3, 3/4$ _____?

- $4/5$

The value of $6! =$

- 720

A constant function is one to one iff its _____ is a singleton.

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- Domain

A constant function is onto iff its _____ is a singleton.

- Co-domain

Number of one to one functions from $X=\{a,b\}$ to $Y=\{u,v\}$ are equal to _____.

- 2

If f is defined recursively by $f(0) = -1$ and $f(n+1)=f(n)+3$, then $f(2)=$ _____.

- 5

A function whose inverse function exists is called a/an _____.

- Invertible function

Let $f(2)=3$, $g(2)=3$, $f(4)=1$ and $g(4)=2$ then the value of $f \circ g(4)$ is.....

- 3

Let $A=\{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B is _____.

- Onto

Composition of a function is a commutative operation.

- False

Composition of a function is not a commutative operation

- True

The sum of first five whole number is _____.

- 10

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

- True

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- $x-2$

The total number of terms in an arithmetic series $0+5+10+15+....50$ are _____.

- 11

$9!/6!=$ _____

- 504

Let $f(x)=3x$ and $g(x)=3x-2$ define functions f and g from R to R , THEN $(F+G)(X)$ is _____

- $6x-2$

If f is a bijective function then $(f \cdot f(x))$ is equal to

- X

A sequence whose terms alternate in sign is called an _____

- Alternating sequence

Common ratio in the sequence "4, 16, 64, 256,..." is.....12.

- 4

0.8181818181 is a infinite geometric series.

- True

The word 'algorithm' refers to a step-by-step method for performing some action.

- True

A predicate become _____ when its variables are given specific values

- Sentence

The sum of two irrational numbers must be an irrational number.

- False

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The sum of two irrational numbers need not be irrational number

- True

The division by zero is allowed in mathematics.

- False

The product of any two consecutive positive integers is divisible by 2

- True

If 'n' is an odd integer then n^3+n is _____.

- Even

For integers a,b,c, If divides b and a divides c, then a divides (a+b).

- False

Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that _____ and $0 \leq r < d$

- $N=d.q+r$

A rule that assigns a numerical value to each outcome in a sample space is called

- Random variable

If A and B are two disjoint (mutually exclusive) events then $P(A \cap B) =$

- $P(A) + P(B)$

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

- $C(10,5)$

The expectation of x is equal to

- $\sum xf(x)$

If $P(A \cap B) = P(A) P(B)$ THEN THE events A and B are called

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- Independent

A walk that starts and ends at the same vertex is called.

- None optins

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

- 497

What is the probability of getting a number greater than 4 when a die is thrown?

- $3/5$

Eater formula for graphs is _____.

- $F=e-v+2$

$X+a, x+3a, x+5a, \dots$ is an _____

- Arithmetic sequence

Composition of a function is a commutative operation.

- False

Real valued function is a function that assigns _____ to each member of its domain.

- Only a real number

Let $X=\{1,2,3,4\}$ and a function 'f' defined on X $f(1)=1, f(2)=2, f(3)=3, f(4)=4$ then _____

- F is an identity function

A constant function is surjective if and only if _____

- The co-domain consists of a single element

Cardinality means the total number of elements in a set.

- True

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If $f(x)=2x$ and $g(x)=x$ then $g(f(x))$ is _____.

- $2x^2$

Let $f:R \rightarrow R$ is one to one function then c, f, c is not equal 0 is also one to one function.

- True

Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1, f(2)=1, f(3)=1, f(4)=1$ then which of the following is true?

- F is a constant function

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

- True

Which of the following is not correct for a 'sequence'?

- A sequence is a relation whose domain is the set of natural numbers

$F(x)=x^2$ is not one to one function from R to R^+

- True

Let $f: R \rightarrow R$ is one to one function then c, f, c is not equal to is also one to one function.

- True

Let $f(x)=x+2$ then $f^{-1}(x)$ is _____

- $x-2$

Let $f(x)=x^2+1$ define functions f from R to R and $c=2$ be any scalar, then $c, f(x)$ is _____.

- $2x^2-1$

One to one correspondence means the condition of _____.

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- Both (a) and (c)

A function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \text{square root } x$ is a real valued function.

- False

If $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = e^x$ is a real valued function of a real variable.

- True

A function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \log x$ is a real valued function

- True

$\frac{1}{2}$, then 3rd term of sequence is _____.

- $\frac{1}{2}$

The process of defining an object in terms of smaller versions of itself is called recursion.

- True

Which of the following is not correct for a 'sequence'?

- A sequence is a relation whose domain is the set of natural numbers

A set is called countable if, and only if, it is _____.

- Finite and countable infinite.....both

A set that is not countable is called _____.

- Uncountable

A sequence whose terms alternate in sign is called an _____.

- Alternating sequence

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is one to one function then $c \circ f$ is also one to one function for _____.

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- C is not equal 0

Let $f(x) = x+3$ then $f^{-1}(x)$ is _____ -

- $x-3$

Let f and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

- $2x+3$

Number of one to one functions from $X=\{a,b\}$ to $Y=\{u,v\}$ are equal to _____

- 2

The fibonacci sequence is defined as $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$ for all integers $k \geq 2$ then which of the following is true for F_2

- $F_2-F_1 = 2+1=3$

$x+a, x+3a, x+5a, \dots$ is an _____.

- Arithmetic sequence

Inverse of a function may not be a function.

- True

In the following sequence $a_k = K/(k+1)$, for $k=1$, a_1 will be _____.

- $\frac{1}{2}$

If $f(x)=x$ and $g(x)=-x$ are both one to one function then $(f+g)(x)$ is also one to one function.

- False

If $f(x)=x$ and $g(x)=-x$ are both one to one function then $(f+g)(x)$ is not one to one function.

- True

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The function 'f' and 'g' are inverse of each other if and only if their composition gives_____.

- Identity function

Which of the following set is the domain of a sequence?

- Set of real numbers

Let C is defined as the set of all countries in the world then C is a _____.

- Finite set

A constant function is surjective if and only if _____.

- The co domain consists of a single element

The sum of the series $a_1 + a_2 + a_3 + \dots$ can be written as _____.

-

Inverse of a function may not be a function.

- True
- 3

Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1, f(2)=1, f(3)=1, f(4)=1$ then which of the following is true?

- F is a constant function

The composition of function is always

- Associative

A set is countably infinite if, and only if, it has the same cardinality as the set of

- Positive integers

Two functions 'f' and 'g' from 'X' to 'Y' are said to be equal if and only if _____.

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• $F(x)=g(x)$ for all 'x' belongs to X
 $Y=x^3$ is a graph of bijective function from R to R.

• True
Domain and range are same for_____.

• Identity function
Let $X=\{1,2,3,4\}$ and a function 'f' defined on X by $f(1)=1, f(2)=2, f(3)=3$

• F is an identity function
Composition of a function is a commutative operation.

• True
Inverse of a surjective function is always a function.

• False
A function whose inverse function exists is called a/an_____.

• Invertible
Given a set X define a function I from X to X by $i(x)=x$ from all x belonging to X . Then_____.

• I is both injective and surjective
Let $f: R \rightarrow R$ is a one to one function then c, f is also one to one function for.

• C is not equal 0
Let $X=\{1,5,9\}$ and $Y=\{3,4,7\}$. Define a function f from X to Y such that $f(1)=7, f(5)=3, f(9)=$ ____. Which is true for f(9) to make it a one-to-one (injective) function?

• 4
Which of the following is not a predecessors of a_k ?

• A_k+1

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Two functions 'f' and 'g' from X to Y are said to be equal if and only if _____.

- $F(x)=g(x)$ for all 'x' belongs to x

The two functions 'f' and 'g' are equal if _____.

- $F(x) = 3x$ and $g(x) = \frac{6x^2+3x}{2x^2+1}$ for all $x \in \mathbb{R}$

If first term of a geometric sequence is 2 and common ratio is $\frac{1}{2}$, then 3rd ter, of sequence.

- $\frac{1}{4}$

$Y = \sqrt{x}$ is an _____ function form $\mathbb{R}^+$ to $\mathbb{R}$

- One to one function

$Y = x^2$ is an _____ function form $\mathbb{R}$ to $\mathbb{R}^+$

- NOT ONE TO ONE FUNTION

If a function $(g \circ f)(x) : X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$, Then the function _____.

- $(g \circ f)$

If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is _____.

- -20

If $f(x) = \sin^{-1}(x)$ and $g(x) = \sin x$ then $g \circ f(x)$ is _____.

- X

$F: X \rightarrow Y$ that is both one to one and onto is called a _____.

- Bijective function

What does 'y' denotes in a geometric sequence?

- Common ratio

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Let g be a function defined by $g(x)=x+1$. Then the composition of $(g \circ g)$.

- $x+2$

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Which of the following is true for the following sequence?

- If n is even, then $C_n=2$ and if n is odd, then $C_n = 0$

The function ' f ' and ' g ' are inverse of each other if and only if their composition gives _____.

- Identity function

$N!$ is defined to be _____.

- The product of the integers from 1 to n

Let $A = \{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B

- Both one to one and onto

A set is called countable if, and only if, it is _____.

- Countably infinite and finite

If $f(x) = x$ and $g(x) = -x$ are both one to one functions then $(f+g)(x)$ is also one to one function.

- False

If a is the 1st term and d be the common difference of an arithmetic sequence then the sequence is $a, a+d, a+2d, a+3d, \dots$

- True

$x+a, x+3a, x+5, \dots$ is a/a_n _____.

- Arithmetic sequence

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inverse of an injective function may not be a function.

- True

$y=x^3$ is a graph of bijective function from $\mathbb{R}$ to $\mathbb{R}$

- False

Two functions 'f' and 'g' from x to y are said to be equal if and only if ____.

- $F(x)=g(x)$ for all 'x' belongs to x

Common ratio in sequence '36, 12, 4, $4/3$,' is

- $1/3$

A set is called finite if, and only if, is the ____ or there is ____.

- Empty set or one-to-one

Let $f(x)=x^2-1$ and $g(x)=x+1$ define functions f and g from $\mathbb{R}$ to $\mathbb{R}$, then $(f/g)(x)$

- $x-1$

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Let g be a function defined by $g(x) = x+1$. Then the composition of $(g \circ g)$.

- $x+2$

$f: X \rightarrow Y$ that is both one to one and onto is called a _____.

- Bijective function

What does 'l' denotes in a geometric sequence?

- Common ratio

Let $A=\{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B is ____.

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- **Both one to one and onto**

- The product of the integers from 1 to n

- **Both b and c**

- $C_n = n/n+1$ for $n > 0$

- **Arithmetic sequence**

- **Arithmetic sequence**

- **Arithmetic sequence**

- **Finite sequence**

- **One-one and ontoboth**

- **True**

- **True**

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- Identity function

A set is called finite if , and only if, it is the _____ or there is _____.

- Empty set or one-to-one

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root } x^2$ for all $x \in R$, then _____.

- $F(x)$ is not equal to $g(x)$

If $f^{-1}(x) = 6 - x/2$ then $f^{-1}(2)$ is _____.

- 2

Let $f(2)=3$, $g(2)=3$, $f(4)=1$ and $g(4)=2$ then the value of $f \circ g(4)$ is _____.

- 3

An important data type in computer programming consists of ____.

- Finite sequence

Let $f(x)=x+3$ then $f^{-1}(x)$ is _____.

- $X-3$

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both onto function. Then $g \circ f: X \rightarrow Z$ is _____.

- One-to-one function

The functions ' f ' and ' g ' are inverse of each other if and only if their composition gives

- Identity function

The two function ' f ' and ' g ' are equal if _____.

- $F(x) = 3x$ and $g(x) = 6x^2 + 3x/2x + 1$ for all $x \in R$

Two functions ' f ' and ' g ' from X To Y are said to be equal if and only if—
—.

- $F(x)$ and $g(x)$ for all ' x ' belongs to X

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Which of the following set is the domain of a sequence?

- **Set of natural numbers**

If 1st term of a geometric sequence is 2 and common ratio is $\frac{1}{2}$, then 3rd term of sequence is _____.

- 1/2

Composition of a function is a commutative operation.

- **True**

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$. Then the function _____ is known as composition of f and g .

- $(g \circ f)$

A set is countably infinite if and only if and only if, it has the same cardinality as the set of _____.

- **Positive integers**

The 3rd term of the sequence $b_n=5^n$ is _____.

- 125

If f is a bijective function then $(f^{-1}(f(x)))$ is equal to ____.

- X

Let $f(x) = x$ and $g(x) = -x$ for all $x \in \mathbb{R}$, then $(f+g)(x)$ is ____.

O

An infinite sequence may have only a finite number of values.

- **True**

The functions fog and gof are always equal .

- **False**

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If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

- True

A function whose range consists of only one element is called _____.

- Constant function

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$. Define a function f from X to Y such that $f(1)=7$, $f(5)=3$, $f(9)=4$ then which of the following statement about ' f ' is true?

- f is both one-to-one and onto

$Y=x^3$ is a graph of bijective function from R to R .

- True

A function whose inverse function exists is called a/an _____.

- Invertible

Let F and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

- $2x + 5$

If r is a positive real number, then the value of r in $3, r, r = -27r$ is

- -9

The _____ of the terms of a sequence forms a series.

- Sum

The sum of first five whole number is _____.

- 10

If $f_k = f_{k-1} + f_{k-2}$ then $f_0 = 1$, $f_1 = 2$, then $f_2 =$ _____.

.3

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Let $f(x) = 3x$ and $g(x) = x + 2$ define functions f and g from R to R , then $(f.g)(x)$ is ____.

- $3x^2 + 6x$

Let R be a relation on a set A . If R is reflexive then its compliment is ____.

- Irreflexive

If $A =$ Set of students of virtual university then A has been written in the ____.

- Descriptive form

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$. Then the function ____ is known as composition of f and g .

- $(g \circ f)$

If X and Y are independent random variables and a and b are constants, then $\text{Var}(aX + bY)$ is equal to

- $a\text{Var}(X) + b\text{Var}(Y)$

Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then number of functions from A to B are ____.

- 8

p is equivalent to q' means ____.

- p is necessary and sufficient for q .

Let A and B be subsets of U with $n(A) = 12$, $n(B) = 15$, $n(A') = 17$, and $n(A \text{ intersection } B) = 8$, then $n(U) =$ ____.

- 29

For the following relation to be a function, x can not be what values?
 $R = \{(2,4), (x,1), (4,2), (5,6)\}$

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- x cannot be 2, 4 or 5

Find the number of the word that can be formed of the letters of the word "ELEVEN".

- 120

There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

- 6

Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

- 16

A predicate becomes _____ when its variables are given specific values.

- statement

Find the number of distinct permutations that can be formed using the letters of the word "BENZENE"

- 420

Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- 40

In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

- 24

If $p \leftrightarrow q$ is True, then _____.

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- p and q both are True.

If A and B be events with $P(A) = 1/3$, $P(B) = 1/4$ and $P(A \cap B) = 1/6$, then $P(A \cup B) = \underline{\hspace{2cm}}$.

- 5/12

an integer n is a perfect square if and only if $\underline{\hspace{2cm}}$ for some integer k.

- $n = k^2$

If A and B are disjoint finite sets then $n(A \cup B) = \underline{\hspace{2cm}}$.

- $n(A) + n(B)$

Let $X = \{2, 4, 5\}$ and $Y = \{1, 2, 4\}$ and R be a relation from X to Y defined by $R = \{(2,4), (4,1), (a,2)\}$. For what value of 'a' the relation R is a function ?

- 5

$\sim(P \rightarrow q)$ is logically equivalent to $\underline{\hspace{2cm}}$.

- $p \wedge \sim q$

A tree is normally constructed from $\underline{\hspace{2cm}}$.

- left to right

A Random variable is also called a $\underline{\hspace{2cm}}$.

- Chance Variable

The conjunction $p \wedge q$ is True when $\underline{\hspace{2cm}}$.

- p is True, q is True

The logical statement $p \wedge q$ means $\underline{\hspace{2cm}}$.

- p AND q

Which of the followings is the factorial form of 5 . 4?

- 5!/3!

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What is the minimum number of students in a class to be sure that two of them are born in the same month?

- 13

If p is false and q is true, then $\sim p \leftrightarrow q$ is _____.

- True

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

- TRUE

$(p \vee \sim p)$ is the _____.

- Tautology

$(-2)! =$ _____ ?

- Undefined

If $p =$ It is raining, $q =$ She will go to college
"It is raining and she will not go to college"
will be denoted by

- $p \wedge \sim q$

Let $X = \{1, 2, 3\}$, then 2-combinations of the 3 elements of the set X are _____?

- $\{1, 2\}, \{1, 3\}$ and $\{2, 3\}$

Let $f(x) = x^2 + 1$ define functions f from R to R and $c = 2$ be any scalar, then $c.f(x)$ is _____.

- $2x^2 + 2$

The disjunction of p and q is written as _____.

- $p \vee q$

If X and Y are independent random variables, then $E(XY)$ is equal to

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- $E(x)E(y)$

How many possible outcomes are there when a fair coin is tossed four times?

- 16

Which of the followings is the product set $A * B * C$? where $A = \{a\}$, $B = \{b\}$, and $C = \{c, d\}$.

- $\{(a, b, c), (a, b, d)\}$

The number of the words that can be formed from the letters of the word, "COMMITTEE" are

- $9! / (2!2!2!)$

One-to-One correspondence means the condition of ____.

- One-One and onto

The functions $f \circ g$ and $g \circ f$ are always equal.

- FALSE

If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- K-Sample

Determine values of x and y , where $(2x, x + y) = (8, 6)$.

- $x = 4$ and $y = 2$

Let g be a function defined by $g(x) = x + 1$. Then the composition of $(g \circ g)(x)$ is ____.

- $x + 2$

What is the truth value of the sentence?

'It rains if and only if there are clouds.'

- False

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Reductio and absurdum' is another name of _____.

- Proof by contradiction

X belongs to A or x belongs to B, therefore x belongs to _____.

- A union B

Which of the followings is the product set $A * B * C$? Where $A = \{a\}$, $B = \{b\}$, and $C = \{c, d\}$.

- $\{(a, b, c), (a, b, d)\}$

Real valued function is a function that assigns _____ to each member of its domain.

- Only a real number

The negation of "Today is Friday" is

- Today is not Friday

A non-zero integer d divides an integer n if and only if there exists an integer k such that _____.

- $n = d k$

The statement $p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$

- True

Let R be the universal relation on a set A then which one of the following statement about R is true?

- R is reflexive, symmetric and transitive.

Let $f(x)=3x$ and $g(x) = 3x - 2$ define functions f and g from R to R. Then $(f+g)(x) =$ _____.

- $6x - 2$

The switches in parallel act just like _____.

- OR gate

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The converse of the conditional statement $p \rightarrow q$ is

- $q \rightarrow p$

If X and Y are random variables, then $E(aX)$ is equal to

- $aE(X)$

Which of the following statements is true according to the Division Algorithm?

- $17 = 5 \times 3 + 2$

Let $p \rightarrow q$ be a conditional statement, then the statement $q \rightarrow p$ is called _____.

- Converse

The disjunction $p \vee q$ is False when _____.

- P is False, q is False.

A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

- 30

The converse of the conditional statement 'If I live in Quetta, then I live in Pakistan' is _____.

- If I live in Pakistan, then I live in Quetta.

The functions 'f' and 'g' are inverse of each other if and only if their composition gives _____.

- Identity function

$P(0, 0) = \underline{\hspace{1cm}}?$

- 1

Let p_1, p_2, p_3 be True premises in a given Truth Table. If the

conjunctions

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of the Conclusion with each of p_1 , p_2 , p_3 are True, then the argument is _____.

- Valid

If p is false and q is false, then $\sim p$ implies q is _____.

- False

A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement?

- 125

Let $A = \{2, 3, 5, 7\}$, $B = \{2, 3, 5, 7, 2\}$, $C =$ Set of first five prime numbers. Then from the following which statement is true ?

- $A = B$

The set of prime numbers is _____.

- Infinite set

The contrapositive of the conditional statement 'If it is Sunday, then I go for shopping' is _____.

- I do Not go for shopping, then it is Not Sunday.

Let p be True and q be True, then $(\sim p \wedge q)$ is _____.

- False

In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

- 30

The method of loop invariants is used to prove _____ of a loop with respect to certain pre and post-conditions.

- correctness

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- 126

If A and B are any two sets, then $A - B = B - A$

- **False**
- There are 5 girls students and 20 boys students in a class. How many students are there in total?**

- 25

None of the above

25

15

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55

48

50

The _____ of the terms of a sequence forms a series.

Sum

Multiplication

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is one to one function then c, f, c is not equal 0 is also one to one function.

False

True

720 =

4!

5!

7!

6!

The composition of functions is always _____.

Onto

Associative

one-to-one

commutative

What will be the fourth term of the following sequence $1/2, 2/3, 3/4, \underline{\hspace{1cm}}$?

5/6

7/8

4/5

6/7

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